Computational Physicist Summer Placement

Harwell, Oxfordshire, United Kingdom

JOB DESCRIPTION

Science and Technology Facilities Council
Salary: £23,493 per annum
Contract Type: Fixed Term - 12 weeks
Hours: Full-time
Location: Rutherford Appleton Laboratory, Harwell, Oxfordshire OX11 0QX
Closing date: 8th March 2026
Interview: Virtually via Microsoft Teams in April 2026
Start date: July 2026

The Particle Physics Department (PPD) in STFC, is one of the largest Particle Physics groups in the UK. We work with people across the world to understand and discover the fundamental building blocks of the Universe. We design, construct and operate some of the most sensitive scientific instruments in the world, collaborating with partners in academia, industry and international laboratories such as CERN. We currently work with the ATLAS, CMS, LHCb, DUNE, Hyper-K, T2K, MICE, Lux-Zeplin, DEAP-3600, CERN DRDs, Electron Ion Collider and AION collaborations.

Due to the high computation costs of particle physics simulation, cuts are often applied to avoid simulating target events which could never have been detected. At the LHCb experiment, these take the form of generator cuts removing all events with tracks outside of the range of the detectors tracking sensors, spanning $\theta \in [10,400]$mrad. Until now, an unaccounted-for source of waste can also be found in the 'synthetic' acceptance regions introduced by LHCb's 4 Tesla dipole magnet. No account is made to dispose of low momentum events swept into unobservable regions by the powerful magnetic field. Since these events serve no purpose in most analyses they could be classified as waste, and if properly modelled, discarded from final simulations all together.

Early estimates suggest that efficiency of simulation within the charm-quark sector could be improved by upwards of 50-60% in the case of 4-body decays. While there is a strong foundation to this work, several open questions remain as to the practicalities of implementing such a cut – and understanding of the mechanical behaviours underlying its structure – the solutions to which would be the aim of this placements research.

Summer Placements at STFC

STFC's incredibly diverse range of placements allow you to work alongside world leading engineers, scientists, and technicians in highly collaborative environments. Whichever you choose, you will become a critical member of the team in which you will be exposed to exciting projects and challenges from the start. You will also be supported every step of the way by our dedicated Early Careers Development Team

Key Duties & Responsibilities

This placement would involve the following:

- Using an existing simplified model, this placement will estimate the losses from this effect for all common applicable particles; the student will be expected to ultimately implement a relevant cut on the generator to address these losses.
- Model the cutoff region in greater detail, to determine more precisely which effects dominate in describing the loss. If an improvement to the existing model is found, you will be expected to update the created cuts accordingly.
- Analyse the behaviours of any unexpected losses to understand their source, and whether it is potentially useful for determining other similar waste-saving cuts. This can be achieved by using a mixture of established detector simulations, toy studies and ultimately mathematical modelling of the region in particle phase space. This will give you familiarity with general programming techniques, and experience implementing and developing for packages/libraries established in HEP research.

Fully supported and with guidance by the team, the successful student would spend their time as an approximately equal split between physics, mathematics, and computer science, and so would be best chosen from at least one of these fields.

Person Specification

The below criteria will be scored during Shortlisting (S), Interview (I) or both (S&I).

In order to apply for the role, we are looking for the following:

Essential:

- You are in full time education at university, and will be returning to university/your studies to complete your undergraduate degree once the summer placement has concluded (S)
- You have the right to live and work in the UK for the duration of the placement (S)
- Be undertaking a degree in the fields of Physics, Mathematics or Computer Science (S)
- Be familiar with the principles of classical Physics and some Relativistic mechanics (S)
- Express confidence in Mathematics, especially Differential & Integral Calculus (S&I)
- Possess moderate proficiency programming in Python (S&I)
- Proficiency in C++ is likely to be useful (S&I)
- Ability to contribute and work well with other team members (S&I)

Employee Benefits

We are offering a fixed term position with a whole host of benefits including:

- 30 days holiday pro rata
- Flexible working hours.
- Easily accessible public transport links/ free parking.
- Excellent learning and development opportunities.

Science and Technology Facilities Council

Link to more benefits: https://stfccareers.co.uk/rewards-and-benefits/

As this job does not fulfil the UK Government minimum criterion for obtaining sponsored migrant worker status, we will be unable to apply for sponsorship for anyone not eligible to work in the UK. At interview, all shortlisted candidates are required to bring with them identification documents and original documents that prove they hold or can obtain the right to work in the UK.

You can check your eligibility here: https://www.gov.uk/check-uk-visa/y